

## Week-8

**a) Write a JSP program to find a factorial of the given number. Fact.jsp**

### **Factorial Calculator**

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF8"%>

<!DOCTYPE html>

<html>

<head>

<title>Factorial Calculator</title>

</head>

<body>

<h2>Factorial Calculator</h2>

<%

String numStr = request.getParameter("number");

if (numStr != null) {

try {

int number = Integer.parseInt(numStr);

if (number < 0) {

out.println("<p style='color:red;'>Please enter a non-negative integer.</p>");

} else {

long factorial = 1;

for (int i = 2; i <= number; i++) {

factorial *= i;

}

out.println("<p>Factorial of <b>" + number + "</b> is <b>" + factorial +
"</b>.</p>");

}

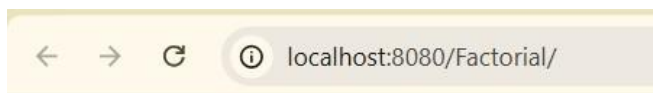
} catch (NumberFormatException e) {
```

```
out.println("<p style='color:red;'>Invalid input. Please enter an integer  
number.</p>");  
}  
}  
%>  
</body>  
</html>
```

### Index.html

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Welcome</title>  
</head>  
<body>  
<form method="post" action="Fact.jsp">  
Enter a number:  
<input type="number" name="number" min="0" required>  
<input type="submit" value=" Calculate">  
</form>  
</body>  
</html>
```

### Output:



A screenshot of a web browser window. The address bar shows 'localhost:8080/Factorial/'. Below the address bar, the text 'Enter a number:' is followed by a text input field containing the number '7'. To the right of the input field is a button labeled 'Calculate'.



A screenshot of a web browser window. The address bar shows 'localhost:8080/Factorial/Fact.jsp'. The page content displays the text 'Factorial of 7 is 5040.'

### Factorial Calculator

Factorial of 7 is **5040**.

**b) Create a user validation web application using JSP, where the user submits the login name and password to the server. The name and password are checked against the data already available in database and if the data matches, a successful login page is returned. Otherwise show a failure message to the user**

MySQL

```
CREATE DATABASE loginDataBase;

USE loginDataBase;

CREATE TABLE users (
id INT AUTO_INCREMENT PRIMARY KEY,
username VARCHAR(50) NOT NULL UNIQUE,
password VARCHAR(50) NOT NULL
);

INSERT INTO users (username, password) VALUES
('divya', 'div123'),
('soumya', 'soum123'),
('ramya', 'ram123');
```

LoginServlet.java

```
package Servlets;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;
import java.sql.*;

@WebServlet("/login")

public class LoginServlet extends HttpServlet {
```

```

private static final String DB_URL = "jdbc:mysql://localhost:3306/loginDataBase";
private static final String DB_USER = "root"; // Replace with your MySQL username
private static final String DB_PASS = "password"; // Replace with your MySQL password

@Override

protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

    request.getRequestDispatcher("login.jsp").forward(request, response);

}

@Override

protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

    String username = request.getParameter("username").trim();
    String password = request.getParameter("password").trim();

    boolean validUser = false;

    try {

        Class.forName("com.mysql.cj.jdbc.Driver");

        try (Connection con = DriverManager.getConnection(DB_URL, DB_USER,
DB_PASS);

        PreparedStatement ps = con.prepareStatement(
"SELECT * FROM users WHERE username=? AND password=?")) {

            ps.setString(1, username);
            ps.setString(2, password);

            try (ResultSet rs = ps.executeQuery()) {

                validUser = rs.next();

            }

        }

    } catch (Exception e) {

        e.printStackTrace();
    }
}

```

```

request.setAttribute("errorMessage", "Database error occurred.");
request.getRequestDispatcher("failure.jsp").forward(request, response);
return;
}

if (validUser) {
    HttpSession session = request.getSession();
    session.setAttribute("username", username);
    request.getRequestDispatcher("success.jsp").forward(request, response);
} else {
    request.setAttribute("errorMessage", "Invalid username or password!");
    request.getRequestDispatcher("failure.jsp").forward(request, response);
}
}
}

```

Login.jsp

```

<!DOCTYPE html>

<html>

<head>

    <title>Login</title>

</head>

<body>

    <h2>User Login</h2>

    <form action="login" method="post">

        Username: <input type="text" name="username" required><br><br>

        Password: <input type="password" name="password" required><br><br>

        <input type="submit" value="Login">

    </form>

</body>

```

```
</html>
```

Success.jsp

```
<!DOCTYPE html>
```

```
<html>
```

```
<head><title>Success</title></head>
```

```
<body>
```

```
<h2 style="color:green;">Login Successful!</h2>
```

```
<p>Welcome to the application.</p>
```

```
</body>
```

```
</html>
```

Failure.jsp

```
<!DOCTYPE html>
```

```
<html>
```

```
<head><title>Login Failed</title></head>
```

```
<body>
```

```
<h2 style="color:red;">Login Failed!</h2>
```

```
<p><%= request.getAttribute("errorMessage") %></p>
```

```
<a href="login.jsp">Try Again</a>
```

```
</body>
```

```
</html>
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<web-app version="6.1" xmlns="https://jakarta.ee/xml/ns/jakartaee"
```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
```

```
https://jakarta.ee/xml/ns/jakartaee/web-app_6_1.xsd">
```

```
<servlet>
```

```
<servlet-name>LoginServlet</servlet-name>
```

```
<servlet-class>Servlets.LoginServlet</servlet-class>

</servlet>

<servlet-mapping>

<servlet-name>LoginServlet</servlet-name>

<url-pattern>/LoginServlet</url-pattern>

</servlet-mapping>

</web-app>
```

### Output:

← → ↻ ⓘ localhost:8080/UserLoginServlet

## User Login

Username:

Password:

**Login Successful!**

Welcome to the application.

← → ↻ ⓘ localhost:8080/

## Login Failed!

Invalid username or password!

[Try Again](#)